Part B Report: Variable Selection in Model-Based Clustering (Poon et al., ICML 2010)

1 Summary of the paper

The paper by Poon et al. addresses variable selection in *unsupervised* clustering. In supervised learning, variable selection has a clear goal (e.g., predict the label); in clustering there are no labels, and the authors argue that real data are often *multifaceted*—the same objects can be meaningfully grouped in several different ways depending on which attributes we emphasise. So instead of *doing* variable selection (choosing one subset and one clustering), they propose to *facilitate* it: discover **multiple facets** (multiple latent variables), each corresponding to a different way of partitioning the data, and present these to a domain expert. The technical contribution is the **Pouch Latent Tree Model (PLTM)**: a rooted tree of discrete latent variables with **pouches** of continuous observed variables at the leaves. Given a latent value, each pouch is modelled as a conditional Gaussian. Parameters are estimated with **EM**, and an **eigenvalue constraint** on the covariances (clipping eigenvalues to $[\sigma^2_{\min}, \gamma \cdot \sigma^2_{\max}]$, with $\gamma = 20$) prevents unbounded likelihood and degenerate solutions. The method is evaluated with **NMI** between recovered latent assignments and ground-truth facet labels on synthetic and real data.

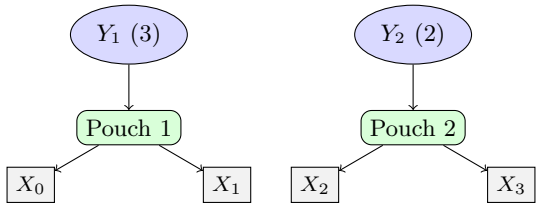


Figure 1: Fixed two-pouch PLTM used in our reproduction: two independent latents Y_1, Y_2; each pouch is a set of observed variables with conditional Gaussian given the parent latent.

2 Reproduction setup and result

We reproduced the **EM algorithm with eigenvalue constraints** on a **fixed** two-facet PLTM (Section 3 of the paper): one latent Y_1 (3 states) for Pouch 1 (features 0–1), one latent Y_2 (2 states) for Pouch 2 (features 2–3), with Y_1 and Y_2 independent. We did not implement structure search or multiple restarts. The dataset is the synthetic multifaceted data from Task 2.1: 1000 samples $\times$ 4 features, with ground-truth labels for Facet 1 (3 clusters on features 0–1) and Facet 2 (2 clusters on features 2–3). We use the same evaluation metric as the paper: **NMI** between the inferred assignments Z_1, Z_2 and the true facet labels.

Our reproduction achieved **NMI(Z_1, Facet 1) $\approx$ 0.91** and **NMI(Z_2, Facet 2) $\approx$ 0.96**. The paper reports NMI ≈ 0.81 (± 0.04) for their synthetic experiment (Table 2, Section 5.3). The gap is an honest reflection of our simplified setup: we used a *fixed* tree and a single run (or few runs), whereas the paper used iterative hill-climbing for structure search and 64 EM restarts. We also used a simpler 4-attribute dataset rather than the paper's 9-attribute setup. Qualitatively, the finding is unchanged: the method recovers multiple facets with strong NMI when the structure matches the data.

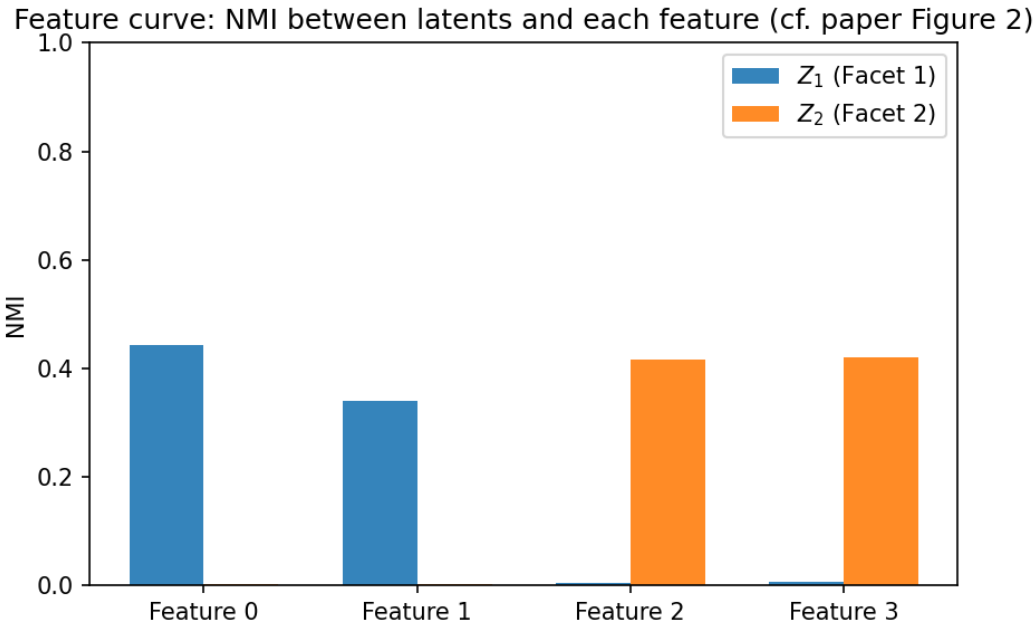


Figure 2: Feature curve (Task 2.3): NMI between each recovered latent (Z_1, Z_2) and each feature. Z_1 specialises on features 0–1 and Z_2 on features 2–3, matching the paper's Figure 2 style.

3 Ablation findings

We ablated two components independently on the same synthetic data and NMI metric.

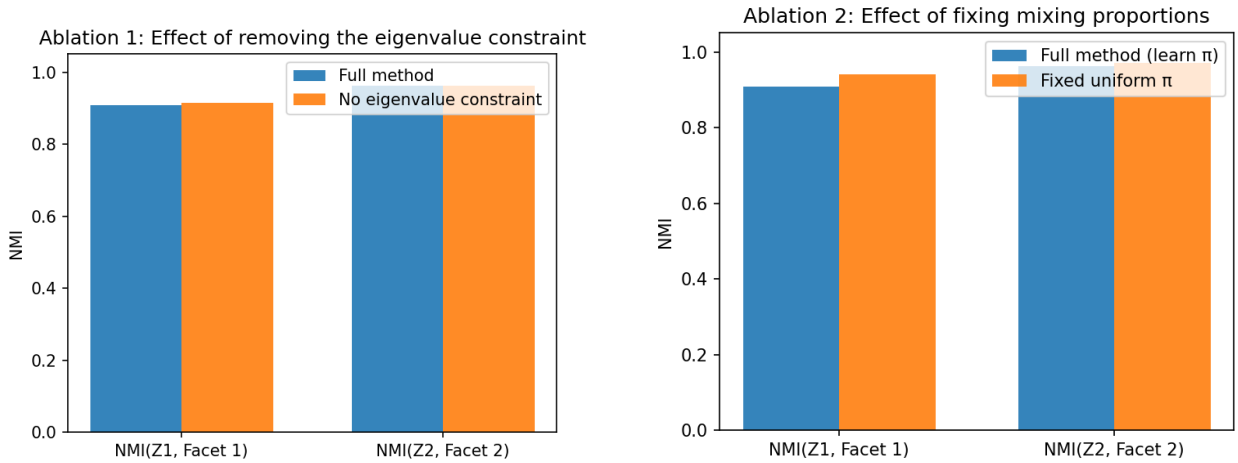
(1) Eigenvalue constraint. The constraint clips each covariance's eigenvalues to $[\sigma^2_{\min}, \gamma \cdot \sigma^2_{\max}]$ after the M-step to avoid degenerate solutions. When we *removed* it (no clipping), NMI for Facet 1 slightly *increased* ($0.9094 \to 0.9156$) and Facet 2 stayed the same (0.9631). On this easy, well-separated toy data, the constraint is not strictly necessary for good recovery; it acts mainly as a **regulariser** for stability. The ablation shows that its main contribution is robustness across harder settings and initialisations, not a large gain on this particular dataset.

(2) Learned mixing proportions π. The full method updates $\mathrm{P}(Y_1)$ and $\mathrm{P}(Y_2)$ from the expected counts in the M-step. We ablated by fixing $\pi_1 = [1/3, 1/3, 1/3]$ and $\pi_2 = [1/2, 1/2]$. On our balanced synthetic data, NMI *improved* slightly (Facet 1: $0.9094 \to 0.9404$; Facet 2: $0.9631 \to 0.9708$) because the true facets are nearly balanced and the fixed uniform π matches that. The ablation shows that **learning** π matters most when facet sizes are imbalanced; when the data are already close to uniform, fixing π can reduce variance in EM and give a small gain.

Together, the ablations indicate that both components are design choices for **stability and adaptability** (eigenvalue constraint) and **adaptation to class balance** (learned π), with effects that are subtle on this toy setup but important in more varied regimes.

4 Failure mode

We constructed a **failure scenario** that violates the **singular parentage** and **tree** assumptions from Task 1.2. In the original setup, each observed variable belongs to exactly one pouch (one latent parent). We generated an **entangled** dataset: we sample two latent vectors u (Facet 1, 3 components) and v (Facet 2, 2 components), then form observed features as a *linear combination* of (u, v) via a full 4×4 mixing matrix, so that *every* observed feature depends on *both* facets. The PLTM is still forced to use two disjoint pouches (features 0–1 vs 2–3), so the model is structurally misspecified. Figure 4 contrasts the assumed tree with



(a) Ablation 1: no eigenvalue constraint.

(b) Ablation 2: fixed uniform π.

Figure 3: Full method vs ablated variants; NMI for Facet 1 and Facet 2.

the true entangled structure.

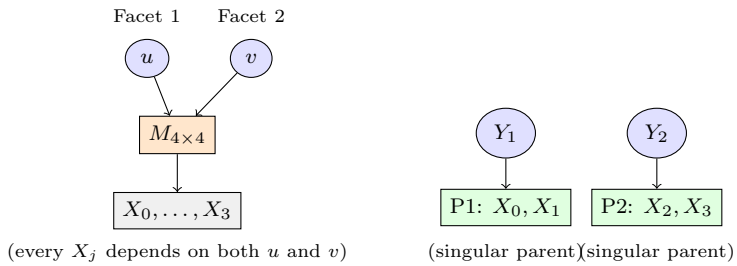


Figure 4: Left: true generative process (entangled). Right: structure assumed by PLTM (tree). Mismatch causes failure on Facet 1.

On this data, **NMI(Z_1, Facet 1) dropped to $\approx$ 0.016** while **NMI(Z_2, Facet 2) could still reach 1.0**. The first pouch (features 0–1) is contaminated by the second facet, so the tree cannot represent the true dependence and recovery of Facet 1 fails. The second pouch can still align with the second latent for this particular mixing matrix, so Facet 2 is recovered. The failure is a direct consequence of violating the assumptions: the model cannot express cross-facet loadings, so when both facets influence the same observed subspace, facet recovery fails. Figure 5 shows the NMI comparison. A natural modification would be to relax the tree to allow **shared latent factors and cross-loadings** (e.g. mixture of factor analyzers) so that a single observed variable can depend on multiple facets.

5 Reflection

We did not implement the full paper: no **structure search** (hill-climbing over tree/pouch configurations), no **64 restarts**, and no experiments on the paper's real datasets (e.g. the 9-attribute synthetic or the applications in Section 5). We focused on a fixed two-pouch PLTM and the EM plus eigenvalue constraint only. It was surprising that removing the eigenvalue constraint or fixing π sometimes *improved* NMI slightly on the toy data—this underscores that these components are about robustness and balance-adaptation rather than always increasing NMI on a single run. The failure-mode experiment clearly showed how one facet (Facet 1) collapses when the data violate singular parentage, while the other can still succeed depending on the mixing. If we had more time, we would add multiple restarts and report mean and standard deviation

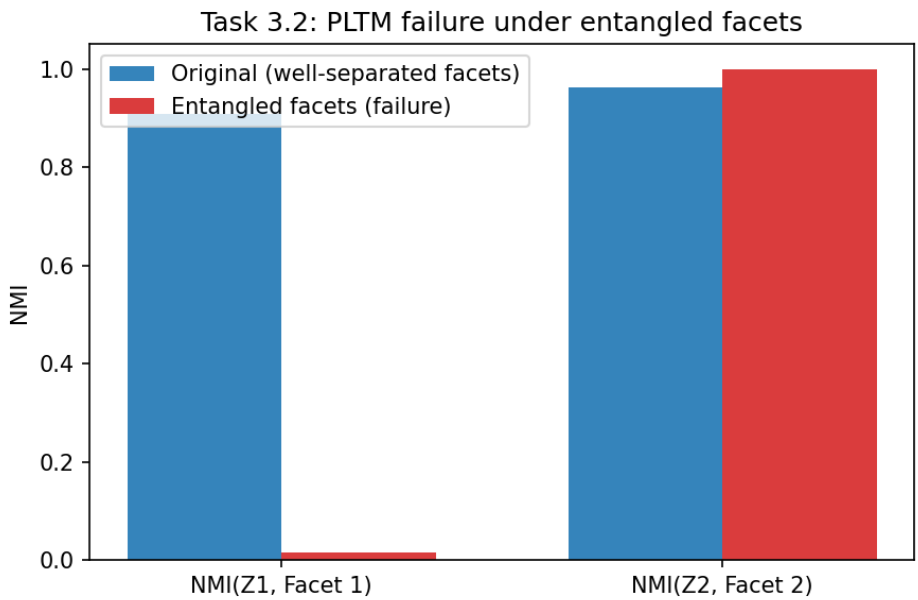


Figure 5: Task 3.2: NMI on original (well-separated facets) vs failure (entangled facets). Facet 1 recovery collapses; Facet 2 can still reach 1.0 for this mixing.

of NMI, run the failure mode with several mixing matrices to see when both facets fail, and implement or compare against the paper's structure-search procedure to close the gap with the reported 0.81 NMI.

References

[1] L. Poon, N. L. Zhang, T. Chen, and Y. Wang. Variable selection in model-based clustering: To do or to facilitate. In *ICML*, 2010.